

Lab Report No 9
Demonstration and practice of delay
generation, loop counter and use of lookup tables.
Embedded Systems



Submitted By:

[Huzaifa Shahbaz]

[21-CE-009]

[Aqsa Shah]

[21-CE-039]

Submitted to:

Engr. Bushra Fiaz

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Department of Computer Engineering,
HITEC University, Taxila

Lab Task No 1:

Solution:

Brief description (3-5 lines)
Implementation of counter 00 to 99 with time duration for each digit being 0.1 seconds on two 7-segment display DS3 and DS4, was done using the design created in task-2 of lab-8 using table lookup.
The code
<pre>; Up Counter Program for 7-Segment Displays using Table Lookup ORG 0000H LJMP MAIN ; Jump to main program ; 7-Segment Display Lookup Table (Common Cathode) ; Each entry represents the pattern to display a digit (0-9) ORG 0030H LOOKUP_TABLE: DB 3FH ; 0 - 0011 1111 DB 06H ; 1 - 0000 0110 DB 5BH ; 2 - 0101 1011 DB 4FH ; 3 - 0100 1111 DB 66H ; 4 - 0110 0110 DB 6DH ; 5 - 0110 1101 DB 7DH ; 6 - 0111 1101 DB 07H ; 7 - 0000 0111 DB 7FH ; 8 - 0111 1111 DB 6FH ; 9 - 0110 1111 ; Main Program ORG 0100H MAIN: ; Initialize Port 1 for output (7-segment displays) MOV P1, #00H ; Clear Port 1 ; Initialize counter variables MOV R2, #00H ; Units digit MOV R3, #00H ; Tens digit COUNTER_LOOP:</pre>

```

; Display current number
ACALL DISPLAY_NUMBER
; Delay for 0.1 seconds
ACALL DELAY
; Increment units digit
INC R2
MOV A, R2
CJNE A, #10, CHECK_CONTINUE
; Reset units digit and increment tens digit
MOV R2, #00H
INC R3
MOV A, R3
CJNE A, #10, CHECK_CONTINUE
; Reset entire counter when reaching 99
MOV R3, #00H
CHECK_CONTINUE:
    SJMP COUNTER_LOOP
; Subroutine to display current number on 7-segment displays
DISPLAY_NUMBER:
    ; Get segment pattern for units digit
    MOV A, R2
    MOVC A, @A+DPTR
    MOV P1, A    ; Display on DS4 (units digit)
    ; Get segment pattern for tens digit
    MOV A, R3
    MOVC A, @A+DPTR
    SWAP A      ; Move tens digit pattern to upper nibble
    ORL P1, A   ; Display on DS3 (tens digit)
    RET
; Delay Subroutine for 0.1 seconds (approximate)
; Assumes 12 MHz crystal oscillator
DELAY:
    MOV R7, #50 ; Outer loop counter
DELAY_OUTER:
    MOV R6, #250 ; Inner loop counter

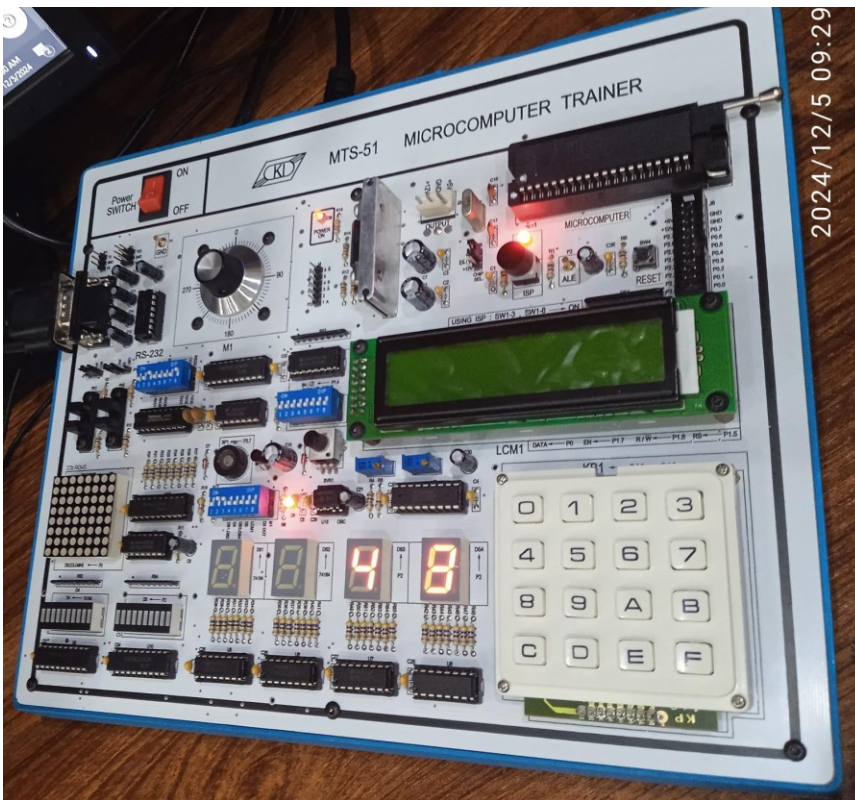
```

```
DELAY_INNER:  
    DJNZ R6, DELAY_INNER  
    DJNZ R7, DELAY_OUTER  
    RET  
END
```

The results (Screenshot)

Build Output

```
creating hex file from ".\Objects\up counter"...\br/>".\Objects\up counter" - 0 Error(s), 3 Warning(s).  
Build Time Elapsed: 00:00:01
```



Lab Task No 2:

Solution:

Brief description (3-5 lines)

Re-write the task 3 of lab 7 using table lookup. This task should be:
Write a program that will show the first alphabet of the name of all

group members on Dot Matrix LED. Each alphabet remains active for a duration of 0.5 seconds.

The code

; 8051 Assembly Program to Display First Alphabets on Dot Matrix LED

; Using Table Lookup Method with Delay and Repetition Technique
ORG 0000H

LJMP MAIN ; Jump to main program

; Dot Matrix LED Column Display Table for First Alphabets
ALPHABET_TABLE:

; 5-column patterns for first alphabets (example patterns)

; Each byte represents one column of the character

; A pattern

DB 07H, 05H, 05H, 05H, 07H

; B pattern

DB 07H, 05H, 07H, 05H, 07H

; S pattern

DB 07H, 04H, 07H, 01H, 07H

; 5ms Delay Subroutine

DELAY_5MS:

MOV R7, #20

DELAY_LOOP1:

MOV R6, #250

DELAY_LOOP2:

DJNZ R6, DELAY_LOOP2

DJNZ R7, DELAY_LOOP1

RET

; Display Character Subroutine

; Input: DPTR points to character pattern, R2 = column counter

DISPLAY_CHAR:

MOV R3, #20 ; Display character 20 times (0.5 seconds)

REPEAT_CHAR:

MOV R2, #5 ; 5 columns to display

COLUMN_LOOP:

; Get column pattern

```

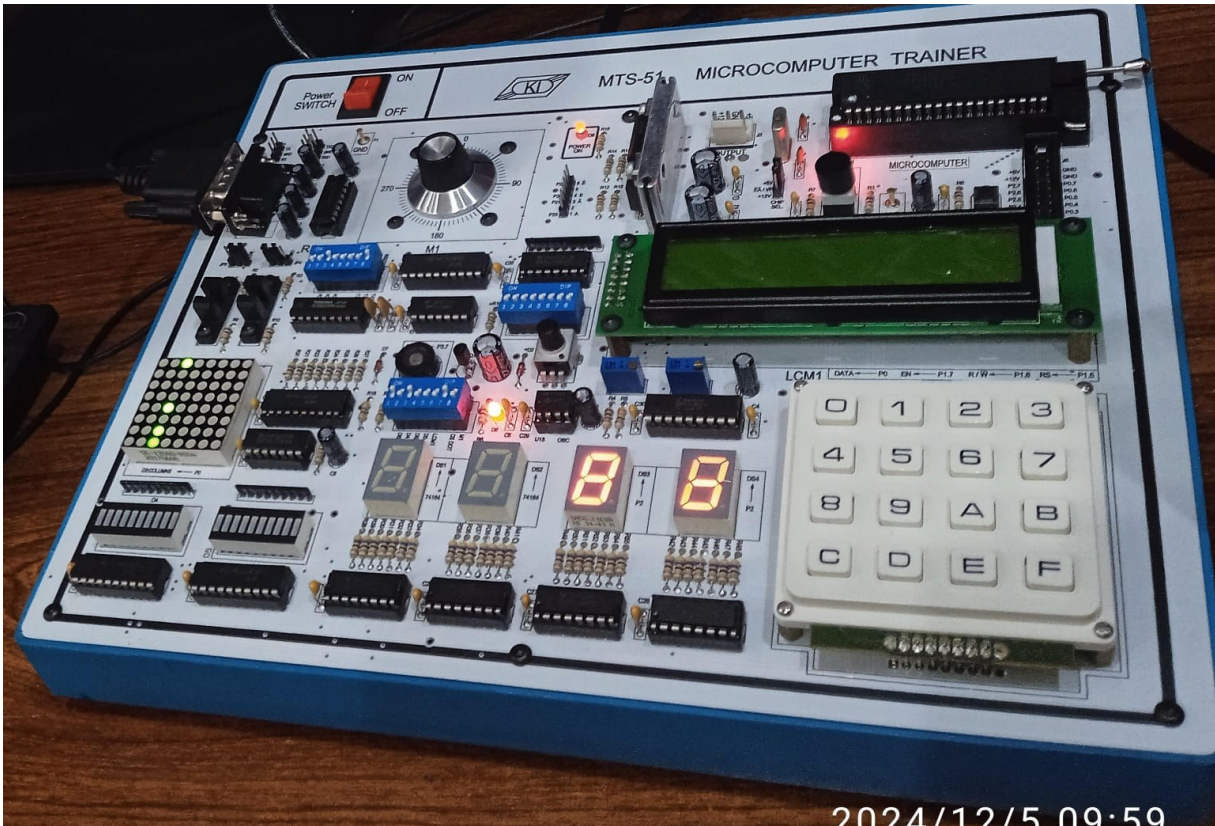
MOV A, R2
DEC A      ; Adjust for 0-based indexing
MOVC A, @A+DPTR ; Get column pattern from table
; Display column
MOV P1, A  ; Output to LED
ACALL DELAY_5MS ; 5ms delay for column display
; Next column
DJNZ R2, COLUMN_LOOP
; Repeat character display
DJNZ R3, REPEAT_CHAR
RET
; Main Program
MAIN:
; Initialize Port
MOV P1, #00H ; Clear Port 1 (Dot Matrix LED)
; Display First Alphabets
; A
MOV DPTR, #ALPHABET_TABLE ; Point to start of table
ACALL DISPLAY_CHAR ; Display 'A'
; B - Move DPTR to B pattern
MOV DPTR, #ALPHABET_TABLE+5 ; Skip 5 bytes to B pattern
ACALL DISPLAY_CHAR ; Display 'B'
; S - Move DPTR to S pattern
MOV DPTR, #ALPHABET_TABLE+10 ; Skip 10 bytes to S
pattern
ACALL DISPLAY_CHAR ; Display 'S'
; Clear display
MOV P1, #00H
; Infinite loop to stop program
STOP:
SJMP STOP
END

```

The results (Screenshot)

Build Output

```
creating hex file from ".\Objects\Dot Matrix LEDLOOKUPTABLE"...  
".\Objects\Dot Matrix LEDLOOKUPTABLE" - 0 Error(s), 3 Warning(s).  
Build Time Elapsed: 00:00:01
```



Conclusion

In conclusion, the above-stated delay generation demonstration and practice with loop counters and lookup tables provide basic insights into microcontroller programming. In real-time applications, accurate timing control is created by means of delay generation techniques. Loop counters reduce the difficulty of performing a repetitive task and increase code execution efficiency. They quickly identify and retrieve data through lookup tables, thus enhancing processing speed and reducing overhead computation. Together, these contribute to performance optimization in embedded systems.